

DANY HAIKAL BIN YAHYA

Firmware Engineer Candidate • Computer & Communication Systems Engineer (Hons.)

Muar, Johor, Malaysia • +60 16-467 2855 • danyhaikal79@gmail.com

Portfolio: danyhaikal79.github.io/Portfolio • GitHub: [@danyhaikal79](https://github.com/danyhaikal79) • LinkedIn: [/in/dany-haikal](https://in.linkedin.com/in/dany-haikal)

PROFESSIONAL SUMMARY

Final-year Computer & Communication Systems Engineering (Hons.) student at Universiti Putra Malaysia with strong foundations in C/C++, Python, embedded systems, and computer architecture. Hands-on ESP32 firmware experience across two projects an RFID inventory system (PN532 over I²C, LCD over SPI, OOP-structured firmware) and a 42-day field-deployed poultry monitoring system (DHT22 polling, UART streaming to a Jetson Nano edge-AI tier). Additional experience with industrial IoT (LoRaWAN gateway + ChirpStack), FPGA design in Verilog using Intel Quartus Prime, and customer-facing technical work through my System Engineer internship at Beyond2U. Available for full-time engineering roles starting 1 September 2026.

TECHNICAL SKILLS

PROGRAMMING C · C++ · Python · Verilog HDL · Java · Bash · SQL

EMBEDDED & FIRMWARE ESP32 (ESP-WROOM-32) · Jetson Nano · PN532 NFC / RFID · ILI9341 LCD · DHT22 · Pi Camera · Arduino framework · OOP-structured firmware

COMMUNICATION PROTOCOLS I²C · SPI · UART · MQTT · Wi-Fi · TCP/IP · ISO/IEC 14443 (NFC/RFID) · LoRaWAN (via ChirpStack + Dragino gateway)

FPGA & HARDWARE Intel Quartus Prime · Verilog HDL · Microprocessor systems · Computer architecture · Digital logic design · PC / Server architecture · BIOS / UEFI

OS & TOOLS Linux (Ubuntu, Debian, Jetson Linux) · Bash / Shell scripting · SSH · Windows admin · Git · GitHub · Docker · Google Colab

AI / ML (APPLIED) PyTorch (YOLOv8 / v11 / v12 / v26 benchmarking) · TensorFlow / Keras (LSTM) · OpenCV · scikit-learn (KNN imputation) · Roboflow

DATA, IoT & STORAGE MySQL · InfluxDB (time-series) · ThingSpeak · Node-RED · Flask REST API · Grafana · Streamlit · ChirpStack · NVMe / SSD fundamentals

ENGINEERING SIMULATION MATLAB · OptiSystem · Multisim

DESIGN & DOCUMENTATION AutoCAD · SolidWorks · Microsoft Visio · Mosquitto MQTT broker

EDUCATION

Universiti Putra Malaysia

Oct 2022 – Sep 2026

Bachelor of Computer and Communication Systems Engineering with Honours

Selangor, Malaysia

- CGPA: 3.43 / 4.00 — Second Upper Class Honours
- Relevant coursework: Microprocessor Technology, Embedded System Design, Computer Architecture, Electronic Communication Circuits, Programming (C/C++/Python), Communication System Design & Simulation, Network Topology Design & Simulation.

Centre of Foundation Studies, UPM

Aug 2021 – Jun 2022

Foundation in Agricultural Science

Selangor, Malaysia

- CGPA: 3.55 / 4.00 — First Class Honours

FEATURED ENGINEERING PROJECTS

RFID Smart Inventory Management System

Embedded Firmware · Full-Stack · ESP32

Tech Stack: ESP32-WROOM · PN532 NFC/RFID (13.56 MHz) · LCD Touchscreen · Embedded C/C++ (Arduino framework) · OOP · I²C · SPI · MySQL

- Led team architecture decisions for a warehouse-grade RFID inventory tracker hardware selection, data-flow design, and UI optimization ensuring the design was executable within hardware constraints while meeting the core project objective.
- Selected hardware: ESP-WROOM-32 microcontroller (dual-core Xtensa LX6, Wi-Fi + Bluetooth), PN532 NFC / RFID reader (ISO/IEC 14443 Type A/B, 13.56 MHz), and an LCD touchscreen chosen for native NFC support and direct visual feedback.
- Developed embedded firmware in C/C++ on the ESP32 (Arduino framework), structuring the code with object-oriented programming to manage button-placement handling, scan-event flow, and on-device data operations cleanly.
- Built the full-stack backend and frontend to handle sort / add / delete / scan operations against an inventory database, with care toward button placement and intuitive data movement on the touchscreen UI.
- Evaluated and tested the finalized device end-to-end to prove the project objective was attained, including failure-mode behavior (lost Wi-Fi, duplicate scans, unknown tags).

Intelligent Poultry & Feed Monitoring System

Embedded Firmware + Edge AI · 42-Day Field Deployment

Tech Stack: ESP32 (Arduino) · DHT22 · UART Serial · Jetson Nano (Linux) · Pi Camera · PyTorch · YOLOv8 / v11 / v12 / v26 · Roboflow · ThingSpeak · Streamlit · SSH

- Planned the full system pipeline and execution schedule using Gantt charts to ensure streamlined development from data-collection to deployment.
- Designed a two-tier embedded architecture: ESP32 microcontroller as a real-time sensor front-end reading DHT22 (temperature + humidity), with the Jetson Nano + Pi Camera handling vision-based feed-level and chicken-count inference.
- Wrote ESP32 firmware in C/C++ (Arduino framework) to poll the DHT22 and stream clean timestamped readings to the Jetson Nano over UART serial resolving the bit-banged-protocol timing issues that occur when DHT22 is connected directly to Linux.
- Deployed the complete data-collection rig at the UPM Serdang study site for 42 continuous days, monitoring system health remotely every 2 days via SSH to detect any hardware or software malfunction.
- Collected and annotated training datasets with Roboflow, then trained and fine-tuned TWO YOLO models — one for chicken counting and one for feed-level estimation benchmarking YOLOv8, YOLOv11, YOLOv12, and YOLO26 to select the best version per task.
- Designed and developed the full IoT system architecture using ThingSpeak (cloud telemetry) and Streamlit (operator dashboard), and validated final model accuracy against a real-world weighing tool to ensure the system performs beyond proof-of-concept.

IoT CO₂ & MoldRisk Prediction System

LoRaWAN · LSTM · Industrial IoT

Tech Stack: Milesight Sensors · Dragino LoRaWAN Gateway · ChirpStack · Node-RED · MQTT · InfluxDB · Python · LSTM · Flask · Grafana · scikit-learn (KNN imputation)

- Deployed and configured an industrial IoT monitoring stack using Milesight environmental sensors, a Dragino LoRaWAN gateway, and ChirpStack network server software handling LoRaWAN device provisioning, network keys, and payload decoding.
- Designed the full data pipeline in Node-RED so that sensor readings (CO₂, pressure, humidity, temperature) flow from ChirpStack → MQTT → InfluxDB for long-term time-series storage.

- Collected sensor data daily through Node-RED for 7 continuous weeks to build a robust dataset for time-series model training.
- Cleaned raw sensor streams using KNN-based imputation in scikit-learn to fill missing values, then engineered rolling-window features for the time-series model.
- Trained an LSTM model in Python to forecast indoor CO₂ levels and MoldRisk 30 minutes ahead, then exposed live inference through a Flask REST API and visualized predictions on a Grafana dashboard.

Stray Dog Detection System (YOLO)

Computer Vision · Real-Time Detection

Tech Stack: YOLOv8 / v11 / v12 · PyTorch · OpenCV · Roboflow · Streamlit · Folium · Google Colab GPU

- Led the team on project planning and task distribution to ensure a streamlined development process.
- Collected, preprocessed, and managed datasets from Kaggle and Roboflow for training and validation.
- Trained and benchmarked multiple YOLO architectures (YOLOv8, YOLOv11, YOLOv12) in PyTorch on Google Colab GPU to select the best-performing model for real-time stray-dog detection.
- Built a Streamlit dashboard with live map visualization (Folium) and alert notifications for traceable monitoring.
- Tested and evaluated the system using online sources for proof of concept validation.

INDUSTRIAL EXPERIENCE

Beyond2U Sdn. Bhd. — System Engineer Intern

Jul 2025 – Oct 2025

Hardware deployment & secure system provisioning

Selangor, Malaysia

- Executed PC tech-refresh and on-site deployment across multiple AmBank branches, troubleshooting hardware and software issues in live production environments.
- Configured IP and domain settings to enforce network isolation and prevent unauthorized external access to bank data.
- Migrated user data from legacy PCs to new machines with zero data loss; installed BitLocker drive encryption with TPM-based key protection.
- Executed multi-pass KillDisk sanitization on decommissioned hardware to eliminate residual data risks.
- Coordinated deployment timing directly with AmBank branch managers and end users, strengthening interpersonal communication and customer-facing technical support skills (directly relevant to firmware RMA troubleshooting).

CERTIFICATIONS & TRAINING

- Wire Bond Course Introduction & Career Talk ; NXP Malaysia, 2024 (semiconductor manufacturing processes, materials, quality control)
- Build Your Own Machine Learning Workshop ; Universiti Putra Malaysia, 2024

LEADERSHIP & COMMUNITY

CSR Project: Unity Through STEM

Pahang, Malaysia · 2024

- Led a team in introducing ferrofluid and STEM concepts to indigenous communities in Taman Negara, Pahang.

CSR Project: Rimba ke Menara Gading

Kuala Lumpur, Malaysia · 2023

- Led members of the Bateq indigenous community on a guided urban exploration of Kuala Lumpur while ensuring their safety throughout the program.

LANGUAGES

- Bahasa Malaysia — Native
- English — Professional Working Proficiency